# [v2.2] Real analysis

## 1. Basics

**Definition 1.1 (Injective)**

A function $f$ is ***injective*** or ***one-to-one*** iff:
- $f(a) = f(b) \Rightarrow a = b$, or equivalently
- $a \neq b \Rightarrow f(a) \neq f(b)$

**Definition 1.2 (Surjective)**

A function $f : A \to B$ is ***surjective*** or ***onto*** iff:
- $f(A) = B$ (its image is its codomain), or equivalently
- for every $b \in B$ there exists $a \in A$ s.t. $f(a) = b$

**Theorem 1.3**

Composition of relations preserves the following properties:
- function
- injective
- surjective
- continuous

## 2. Fields and order

### 2.1. Fields

**Definition 2.1.1 (Field)**

A ***field*** $\mathbb{F}$ is any set with relations $+$, $*$, and $=$ which satisfies
- If $x$ and $y$ are in $\mathbb{F}$, so are both $x + y$ and $xy$.
- $x + y = y + x$ and $xy = yx$.
- $(x + y) + z = x + (y + z)$ and $(xy)z = x(yz)$.
- $x(y + z) = xy + xz$.
- $w = x$ and $y + z \implies w + y = x + z$ and $wy = xz$.
- Given $x, y \in \mathbb{F}$, there exists $z \in \mathbb{F}$ s.t. $x + z = y$ (i.e. all addition problems have solutions).
- There exists some number in $\mathbb{F}$ which is not equal to 0. If $x, y \in \mathbb{F}$ s.t. $x \neq 0$, then there exists a real number $z$ s.t. $xz = y$. This $z$ is denoted $(y/x)$.
- $=$ is an equivalence relation.

### 2.2. Order

**Definition 2.2.1 (Order)**

A relation $<$ that satisfies the [total] order axioms:
- **Trichotomy:** $x < y$, $x = y$, xor $x > y$.
- **Transitivity**: If $x < y$ and $y < z$, then $x < z$.

**Definition 2.2.2 (Dedekind completeness)**

An ordered set is ***complete***, ***Dedekind-complete***, or has the ***least upper bound property*** iff:
- every subset of the set that has an upper bound has a supremum, or equivalently
- every subset of the set that has a lower bound has an infimum.

### 2.3. Ordered fields

**Definition 2.3.1 (Ordered field)**

A field with an order satisfying
- **Transl.:** If $x < y$, then $x + z < y + z$ for all $z$.
- **Pos. closure:** If $x > 0$ and $y > 0$, then $xy > 0$.

### 2.4. Real numbers

**Theorem 2.4.1**

There is a unique complete ordered field, called $\mathbb{R}$, the ***real numbers***.

**Theorem 2.4.2 (Approximation Property)**

Let $S$ be a nonempty set of reals with a supremum. Then for every real number $a < \sup S$ there exists $x \in S$ with $a < x \leq \sup S$.

**Theorem 2.4.3 (Additive Property)**

Let $A$ and $B$ be sets of reals with suprema. Consider

$$C := \{a + b : a \in A, b \in B\}.$$

Then,

$$\sup C = \sup A + \sup B.$$

**Theorem 2.4.4 (Comparison Property)**

Let $S$ and $T$ be two nonempty sets of reals where $s \leq t$ for all $s \in S, t \in T$. Then, if $T$ has a supremum, $S$ also has a supremum and $\sup S \leq \sup T$.

**Theorem 2.4.5 (Archimedean property)**

The reals are comensural: for any $x, y \in \mathbb{R}$ with $x > 0$, then there is a positive integer $n$ s.t. $nx > y$.

### 2.5. Topology of the reals

**Definition 2.5.1 (Open set)**

The following are ***open***:
- The empty set
- Open intervals
- Unions of open sets
- Finite intersections of open sets
- The whole space

A set is ***closed*** if it is the complement of an open set.

The collection of closed sets is closed under finite union and arbitrary intersection.

**Definition 2.5.2 (Continuity)**

Let $f : X \to Y$ be a function from topological space $X$ to topological space $Y$. Then, $f$ is continuous iff $f^{-1}(A) \subset X \Leftrightarrow A$ is open in $Y$.

## 2.6. Absolute values

**Theorem 2.6.1 (Corollaries of the Triangle Inequality)**

For any $x, y \in \mathbb{R}$:

$$|x + y| \leq |x| + |y|$$
$$|x - y| \leq |x| + |y|$$
$$||x| + |y|| \leq |x - y|$$

# 3. Sequences

**Definition 3.1 (Limit of a sequence)**

Let $\{x_n\}$ be a sequence. $L$ is the ***limit***, denoted $\lim_{n\to\infty} x_n$, of the sequence iff for every $\varepsilon > 0$ there exists $N$ s.t. whenever $n > N$, $|x_n - L| < \varepsilon$.

Note - monotonic does not require strictly increasing/ decreasing, it can be stable at some points.

**Theorem 3.2**

A bounded monotonically increasing sequence converges to its supremum.

A bounded monotonically decreasing sequence converges to its infimum.

**Theorem 3.3**

Every nonempty bounded subset of the reals has an increasing sequence converging to its supremum and a decreasing sequence converging to its infimum.

## 3.1. Subsequences

**Theorem 3.1.1**

The following are equivalent:
1. The sequence $\{x_n\}$ converges.
2. The $t$-tail $\{x_{n+t}\}$ converges for every $t$.
3. There is some $t$ for which the $t$-tail converges.

## 3.2. Limit properties and theorems

**Theorem 3.2.1**

Let $\{x_n\} \to x$ and $\{y_n\} \to y$, s.t. $x_n \leq y_n$ for all $n$. Then, $x \leq y$.

**Theorem 3.2.2 (Squeeze Theorem)**

Let $\{a_n\}$, $\{b_n\}$, and $\{c_n\}$ be sequences s.t. $a_n \leq b_n \leq c_n$ for all $n$ and $a_n \to L$ and $c_n \to L$. Then $b_n \to L$.

**Theorem 3.2.3**

Let $\{x_n\} \to x$ and $\{y_n\} \to y$. Then
1. $\{x_n + y_n\} \to x + y$.
2. $\{x_n - y_n\} \to x - y$.
3. $\{x_n y_n\} \to xy$.
4. $\{x_n / y_n\} \to x/y$ if $y_n \neq 0$ and $y \neq 0$ for all $n$.

**Theorem 3.2.4**

Let $\{x_n\}$ be a sequence and $x$ a number. Then,

$$\{x_n\} \to x \iff \{x_n - x\} \to 0.$$

**Theorem 3.2.5 (Convergence of geometric series)**

Let $c > 0$. Then,
1. $c < 1 \Longrightarrow \lim\limits_{n\to\infty} c^n = 0$.
2. $c > 1 \Longrightarrow \{c^n\}$ is unbounded.

**Theorem 3.2.6 (Ratio Test)**

Let $\{x_n\}$ be an always-nonzero sequence s.t.

$$\lim_{n\to\infty} \frac{|x_{n+1}|}{|x_n|} = L.$$

Then,
- $L < 1 \Longrightarrow \{x_n\} \to 0$.
- $L > 1 \Longrightarrow \{x_n\}$ is unbounded.

**Theorem 3.2.7**

$$\lim_{n\to\infty} n^{\frac{1}{n}} = 1$$

## 3.3. Limsup and liminf

**Definition 3.3.1 (Limsup and liminf)**

Let $\{x_n\}$ be a bounded sequence. For each $n$, define

$$a_n := \sup\{x_k : k \geq n\}$$
$$b_n := \inf\{x_k : k \geq n\}$$

Thus, $a_n$ is above the $n$-tail of the sequence, while $b_n$ is below it.

Define

$$\limsup_{n\to\infty} x_n = \lim_{n\to\infty} a_n$$
$$\liminf_{n\to\infty} x_n = \lim_{n\to\infty} b_n.$$

We know that $\limsup x_n \geq \liminf x_n$.

**Theorem 3.3.2**

If $\{x_n\}$ is bounded, then:
- there is a subsequence $\left\{x_{n_k}\right\} \to \limsup x_n$.
- there is a subsequence $\left\{x_{m_k}\right\} \to \liminf x_n$.

**Theorem 3.3.3**

Let $\{x_n\}$ be a bounded sequence with $\limsup x_n = \liminf x_n = L$. Then, $\{x_n\} \to L$.

**Theorem 3.3.4**

$\{x_n\} \to L$ iff every subsequence converges to $L$.

## 3.4. Bolzano-Weierstrass

**Theorem 3.4.1 (Bolzano-Weierstrass for Sequences)**

Every bounded sequence of real numbers (or in $\mathbb{R}^n$) has a convergent subsequence.

In topology terms, a subset of $\mathbb{R}^n$ is sequentially compact iff it is closed and bounded.

**Definition 3.4.2 (Open ball)**

Let $x \in \mathbb{R}^n$. The ball $B(x, \varepsilon)$ of radius $\varepsilon$ about $x$ is defined as

$$B(x, \varepsilon) := \{y \in \mathbb{R}^n : |x - y| < r\}.$$

**Theorem 3.4.3 (Bolzano-Weierstrass theorem)**

Every bounded infinite subset of $\mathbb{R}^n$ has an accumulation point.

**Theorem 3.4.4 (Cantor Intersection Theorem)**

Let $S_1, S_2, \ldots$ be a sequence of bounded closed subsets of $\mathbb{R}^n$ nested as follows: $S_1 \supset S_2 \supset \cdots$. Then, the intersection $S_1 \cap S_2 \cap \cdots$ is closed and nonempty.

### 3.5. Cauchy sequences

**Definition 3.5.1 (Cauchy sequence)**

A sequence $\{x_n\}$ is called ***Cauchy sequence*** iff for every $\varepsilon > 0$, there exists $N$ s.t. for all $m, n > n$, $|x_n - x_m| < \varepsilon$.

In short, beyond a certain term, all the following terms are arbitrarily close.

Every Cauchy sequence is bounded. A sequence of reals converges iff it is a Cauchy sequence.

## 4. Series

### 4.1. Basic convergence tests

**Theorem 4.1.1 ($n^{\text{th}}$ term test for divergence)**

If $\sum x_n$ converges, then $x_n \to 0$.

Equivalently, if $x_n \nrightarrow 0$, $\sum x_n$ diverges.

**Theorem 4.1.2 (Linearity of series)**

Let $\sum x_n = x$ and $\sum y_n = y$. Then, $\sum(x_n + y_n) = x + y$.

For any $c \in \mathbb{R}$, $\sum(cx_n) = cx$.

**Theorem 4.1.3 (p-series)**

$\sum_{n=1}^{\infty} \frac{1}{n^p}$ converges iff $p > 1$.

### 4.2. Absolute convergence

**Lemma 4.2.1**

If $x_n \geq 0$ for all $n$, then $\sum x_n$ converges iff the sequence of partial sums is bounded above. Furthermore, each partial sum is less than or equal to the sum.

**Definition 4.2.2 (Absolute convergence)**

Iff $\sum |x_n|$ converges, then $\sum x_n$ ***absolutely converges***. If $\sum |x_n|$ diverges, but $\sum x_n$ converges, then $\sum x_n$ ***conditionally converges***.

**Theorem 4.2.3 (Comparison test for positive series)**

Let $\sum x_n$ and $\sum y_n$ be series with $0 \leq x_n \leq y_n$ for all $n$. Then:

1. If $\sum y_n$ converges, then $\sum x_n \leq \sum y_n$ ($\sum x_n$ converges).
2. If $\sum x_n$ diverges, so does $\sum y_n$.

### 4.3. Ratio test and root test

**Theorem 4.3.1 (Ratio test)**

Let $\sum x_n$ be a series of nonzero terms. Suppose that

$$L := \lim_{n \to \infty} \frac{|x_{n+1}|}{|x_n|}$$

exists. Then,

- If $L < 1$, $\sum x_n$ converges absolutely.
- If $L > 1$, $\sum x_n$ diverges.
- If $L = 1$, the test gives no information.

**Theorem 4.3.2 (Root test)**

Let $\sum x_n$ be a series and $L := \limsup \sqrt[n]{|x_n|}$ (which may be infinite). Then,

- If $L < 1$, $\sum x_n$ converges absolutely.
- If $L > 1$, $\sum x_n$ diverges.
- If $L = 1$, the test gives no information.

### 4.4. Alternating series

**Theorem 4.4.1 (Alternating series test)**

Let $x_n \to 0$ be monotone decreasing. Then, $\sum (-1)^n x_n$ converges.

**Theorem 4.4.2**

Let $\sum x_n$ be an absolutely convergent series converging to $L$. Let $\sigma : \mathbb{N} \to \mathbb{N}$ be a bijection. Then, $\sum x_{\sigma n}$ is also absolutely convergent and converges to $L$.

In short, if the series is absolutely convergent, we can rearrange the terms and the sum is the same.

**Theorem 4.4.3 (Alternating Series Rearrangement Theorem)**

Let $\sum x_n$ be a series that satisfies the alternating series test but which is not absolutely convergent. Then, for any $L \in \mathbb{R}$ there is a rearrangement of the series that sums to $L$.

### 4.5. Multiplying series

**Definition 4.5.1 (Cauchy product)**

The ***Cauchy product*** of $\sum a_n$ and $\sum b_n$ is defined as

$$\sum_{k=1}^{n} a_n b_{n+1-k}.$$

Essentially, we collect the terms of the product of $\sum a_n$ with $\sum b_n$ in diagonal strips when the terms are arranged in a grid.

**Theorem 4.5.2 (Mertens' theorem)**

Let $\sum a_n \to A$ and $\sum b_n \to B$. Then if at least one of these is absolutely convergent, then $\sum c_n \to AB$, where $\sum c_n$ is the Cauchy product of $\sum a_n$ and $\sum b_n$.

Additionally, if both converge absolutely, so does the sum of $c_n$s.

## 4.6. Power series

**Definition 4.6.1 (Power series)**

Given $x_0 \in \mathbb{R}$ and a sequence $a_n$, we can create the ***power series***, a function of $x$:

$$\sum_{n=0}^{\infty} a_n (x - x_0)^n.$$

A power series ***converges*** iff there exists $x \neq x_0$ that makes the series converge. If $x = x_0$ makes the series converge, then the series converges for any $x$.

**Theorem 4.6.2**

A power series either diverges for all $x \neq x_0$, converges absolutely for all $x$, or there exists a real number $\rho$, called the ***radius of convergence***, s.t. the series converges absolutely on $(x_0 - \rho, x_0 + \rho)$ and diverges if $|x - x_0| > \rho$.

When $x = x_0 + \rho$ or $x = x_0 - \rho$, this theorem is inconclusive.

# 5. Limits and continuity

## 5.1. Limits

**Definition 5.1.1 (Accumulation point)**

$x$ is a ***accumulation point*** of $S$ iff

- every ball $B(x, \varepsilon)$ contains a point of $S$ other than $x$, or, equivalently,
- there is a sequence of points in $S - \{x\}$ converging to $x$.

**Definition 5.1.2 (Limit)**

Let $S$ be a subset of $\mathbb{R}$ and $c$ be an accumulation point of $S$. Let $f : S \to \mathbb{R}$ be a function.

The ***limit*** of $f(x)$ as $x \to c$ is $L$ iff for each $\varepsilon > 0$, there exists a $\delta > 0$ s.t. whenever $x \in S - \{c\}$ and $|x - c| < \delta$, then $|f(x) - L| < \varepsilon$.

**Theorem 5.1.3**

Let $S$ be a subset of $\mathbb{R}$ and $c$ be an accumulation point of $S$. Let $f : S \to \mathbb{R}$ be a function.

Then $f(x) \to L$ as $x \to c$ iff for every sequence $\{x_n\}$ in $S - \{c\}$ whose limit is $c$, the sequence $\{f(x_n)\} \to L$.

## 5.2. Continuity

**Definition 5.2.1 (Continuity)**

Let $S \subset \mathbb{R}$, $c \in S$, and $f : S \to \mathbb{R}$ be a function. Then, $f$ is ***continuous*** at $c$ iff for every $\varepsilon > 0$, there is a $\delta > 0$ s.t. whenever $|x - c| < \delta$, then $|f(x) - f(c)| < \varepsilon$.

**Theorem 5.2.2**

Let $S \in \mathbb{R}$ and $f : S \to \mathbb{R}$ be a function. Let $c$ be a point in $S$. Then:

1. If $c$ is not an accumulation point of $S$, then $f$ is continuous at $c$.
2. If $c$ is an accumulation point of $S$, then $f$ is continuous at $c$ iff $\lim_{x \to c} f(x) = f(c)$.
3. $f$ is continuous at $c$ iff for every sequence $\{x_n\} \in S$ with limit $c$, $f(x_n) \to f(c)$.

## 5.3. Big Continuity Theorems

**Definition 5.3.1 (Bounded function)**

A function $f : S \to \mathbb{R}$ is bounded iff its image is bounded.

**Lemma 5.3.2**

Let $[a, b]$ be a closed finite interval, and $f$ continuous on $[a, b]$. Then $f$ is bounded.

**Theorem 5.3.3 (Extreme Value Theorem)**

Let $f(x)$ be a continuous function defined on a closed interval $[a, b]$. Then, $f$ achieves an absolute maximum and minimum on $[a, b]$.

**Lemma 5.3.4**

Let $f(x)$ be continuous on the closed bounded interval $[a, b]$. If $f(a) < 0$ and $f(b) > 0$, then there exists $c \in [a, b]$ s.t. $f(c) = 0$.

**Theorem 5.3.5 (Intermediate Value Theorem)**

Let $f$ be continuous on closed, bounded $[a, b]$. Let $y \in \mathbb{R}$ s.t. $f(a) < y < f(b)$ or $f(a) > y > f(b)$. Then, there exists $c \in (a, b)$ with $f(c) = y$.

## 5.4. Uniform continuity

**Definition 5.4.1 (Uniform continuity)**

Let $S \subset \mathbb{R}$ and $f : S \to \mathbb{R}$ be a function. For any $\varepsilon > 0$, $f$ is ***uniformly continuous*** on $S$ iff we can find $\delta$ s.t. for all $x, c \in S$ with $|x - c| < \delta$, then $|f(x) - f(c)| < \varepsilon$.

This differs from continuity iff the same $\delta$ works for every point in $S$.

Uniform continuity obviously implies continuity.

**Theorem 5.4.2**

If $f : [a, b] \to \mathbb{R}$ is a continuous function defined on a closed bounded interval, it is uniformly continuous.

**Lemma 5.4.3**

If $f : S \to \mathbb{R}$ is uniformly continuous and $\{x_n\}$ is a Cauchy sequence in $S$, then $\{f(x_n)\}$ is a Cauchy sequence.

**Theorem 5.4.4**

Let $f : (a, b) \to \mathbb{R}$. Then $f$ is uniformly continuous iff it has a right-handed limit at $a$, a left-handed limit at $b$, and the extension of $f$ to the closed interval $[a, b]$ is continuous.

## 5.5. Lipschitz continuity

**Definition 5.5.1 (Lipschitz continuity)**

Let $f : S \to \mathbb{R}$ be a function. If there exists $K \in \mathbb{R}$ s.t. for every $x, y \in S$, $|f(x) - f(y)| < K|x - y|$ then $f$ is ***Lipschitz continuous***.

**Theorem 5.5.2**

Lipschitz continuity implies uniform continuity.

# 6. Derivatives

**Definition 6.1 (Derivative)**

Let $f(x)$ be a real-valued function defined on an open interval $(a, b)$, and let $x_0 \in (a, b)$. The ***derivative*** of $f$ at $x_0$ is

$$\frac{df}{dx}(x_0) := f'(x_0) := \lim_{x \to x_0} \frac{f(x) - f(x_0)}{x - x_0}.$$

Iff this limit is defined, the function is ***differentiable*** at $x_0$.

**Theorem 6.2 (Continuity theorem)**

Let $f$ be defined on $(a, b)$ with $x_0 \in (a, b)$. Then, $f$ is differentiable at $x_0$ iff there is a function $f^*$ (which depends on $f$ and on $x_0$) s.t.

$$f(x) = f(x_0) + f^*(x)(x - x_0)$$

with $f^*$ defined on $(a, b)$ and continuous at $x_0$, with $f^*(x_0) = f'(x_0)$.

If $f$ is continuous, $f^*$ can be defined as

$$f^*(x) := \begin{cases} \frac{f(x) - f(x_0)}{x - x_0} & \text{if } x \neq x_0 \\ f'(x_0) & \text{if } x = x_0. \end{cases}$$

**Theorem 6.3 (Sum, product, and quotient rules)**

Let $f$ and $g$ be functions defined on an interval $(a, b)$, both differentiable at the point $x_0 \in (a, b)$. (In particular, the derivatives there cannot be infinite.) Then, the following derivatives exist and take on the following values:

- $(f \pm g)'(x_0) = f'(x_0) \pm g'(x_0)$.
- $(f * g)'(x_0) = f(x_0)g'(x_0) + f'(x_0)g(x_0)$.
- $(f/g)'(x_0) = \dfrac{g(x_0)f'(x_0) - f(x_0)g'(x_0)}{g(x_0)^2}$.

**Theorem 6.4 (Chain rule)**

Let $f : (a, b) \to \mathbb{R}$ and $g : f(a, b) \to \mathbb{R}$. Let $x_0 \in (a, b)$ with $f$ differentiable at $x_0$ and $f(x_0)$ contained in the interior of $f(a, b)$, and let $g$ be differentiable at $f(x_0)$. Then, $(g \circ f)(x)$ is differentiable at $x_0$, and

$$(g \circ f)'(x_0) = g'(f(x_0))f'(x_0),$$

or equivalently,

$$\left.\frac{dg}{dx}\right|_{x_0} = \left.\frac{dg}{df}\right|_{f(x_0)} * \left.\frac{df}{dx}\right|_{x_0}.$$

## 6.1. Mean value theorem

**Theorem 6.1.1**

Let $f'(c) > 0$ and $f$ continuous in some interval about $c$. Then, there is a $\delta > 0$ s.t.:

- $c < x < c + \delta \implies f(c) < f(x)$
- $c - d < x - c \implies f(x) < f(c)$

Equivalently, if the derivative is positive, then the function is increasing at that point.

Similarly, if the derivative is negative, then the function is increasing at that point.

**Corollary 6.1.1.1**

If $f(c)$ is a local min or local max, than $f'(c) = 0$.

**Definition 6.1.2 (Critical point)**

A point where the derivative of the function is zero.

**Theorem 6.1.3 (Rolle's theorem)**

Let $f(x)$ be continuous on $[a, b]$ and differentiable on $(a, b)$, s.t. $f(a) = f(b)$. Then, there exists $c \in (a, b)$ for which $f'(c) = 0$.

**Theorem 6.1.4 (Mean value theorem)**

Let $f$ be continuous on $[a, b]$ and differentiable on $(a, b)$. Then, there exists some point $c \in (a, b)$ for which

$$f'(c) = \frac{f(b) - f(a)}{b - a}.$$

**Theorem 6.1.5 (Cauchy's mean value theorem)**

Let $f(t)$ and $g(t)$ be two functions continuous on $[a, b]$ and differentiable on $(a, b)$. Then, there exists some point $c \in (a, b)$ for which

$$f'(c)(g(b) - g(a)) = g'(c)(f(b) - f(a)).$$

**Theorem 6.1.6**

Let $f(x)$ be continuous on $[a, b]$ and differentiable on $(a, b)$. Then:

- $f'(x) > 0$ for all $x \in (a, b) \iff f$ is strictly increasing
- $f'(x) < 0$ for all $x \in (a, b) \iff f$ is strictly decreasing
- $f'(x) = 0$ for all $x \in (a, b) \iff f$ is constant

Let $g(x)$ also be continuous on $[a, b]$ and differentiable on $(a, b)$. Then:

- $g'(x) = f'(x)$ for all $x \in (a, b) \iff f(x) - g(x)$ is constant

**Theorem 6.1.7**

Let $f(x)$ be continuous on $(a, b]$ and differentiable on $(a, b)$, with $\lim_{x \to b^-} f'(x) = L$. Then, $f$ is left-hand differentiable at $b$, and $f'_-(b) = L$.

A similar theorem exists for right-hand derivatives.

**Theorem 6.1.8 (Darboux theorem)**

Let $f(x)$ be a function that is differentiable on $[a, b]$. Let $y$ be a value s.t. $f'(a) < y < f'(b)$ or $f'(a) > y > f'(b)$. Then there is some $c \in (a, b)$ s.t. $f'(c) = y$.

**Corollary 6.1.8.1**

- $f'$ cannot change signs without going through zero.
- If $f'(x) \neq 0$ for any $x$, then $f$ is monotonic.
- If $f'$ itself is monotonic then it must be continuous:
  - the only kinds of discontinuities of monotonic functions are jumps
  - derivatives can't make jumps

## 6.2. Taylor series

**Theorem 6.2.1 (Taylor's formula with the Lagrange form of the remainder)**

Let $f : (a, b) \to \mathbb{R}$ have $n + 1$ derivatives on the interval $(a, b)$, and let $x_0$ be a point in $(a, b)$. Then, for any point $y$ in $(a, b)$ ***Taylor's formula*** states that

$$f(y) = p_n(y) + R_n(y; x_0).$$

We define the ***$n$th Taylor polynomial centered at $x_0$*** as:

$$p_n(x) := \sum_{k=0}^{n} \frac{f^k(x_0)}{k!}(x - x_0)^k.$$

The ***Taylor series*** is the limit of the Taylor polynomials.

We define the ***Langrange form of the remainder*** as:

$$R_{n(x;x_0)} := \frac{f^{n+1}(c)(y - x_0)^{n+1}}{(n + 1)!}$$

for some $c$ between $y$ and $x_0$ [this theorem does not specify which $c$ this is].

# 7. Integrals

## 7.1. Partitions

**Definition 7.1.1 (Partition)**

A ***partition*** $P$ of $[a, b]$ is a finite set of points, $\{x_0, x_1, ..., x_n\}$ with $a = x_0 < x_1 < \cdots < x_n = b$.

The $k$th interval of the partition is $[x_{k-1}, x_k]$, with width $\Delta x_k := x_k - x_{k-1}$.

**Definition 7.1.2 (Norm/mesh)**

The ***norm*** or ***mesh*** of a partition is the width of its largest interval, denoted

$$\|P\| := \max\{\Delta x_k : k \leq n\}.$$

**Definition 7.1.3 (Refinement)**

$\tilde{P}$ is a ***refinement*** of $P$ iff $P \subseteq \tilde{P}$.

## 7.2. Darboux integral

**Definition 7.2.1 (Darboux sum)**

Let $f : [a, b] \to \mathbb{R}$ be bounded and $P$ a partition of $[a, b]$. Then, define

$$m_k := \inf\{f(x) : x_{k-1} \leq x \leq x_k\}$$
$$M_k := \sup\{f(x) : x_{k-1} \leq x \leq x_k\}.$$

Define the ***upper*** and ***lower Darboux sums***, respectively, as

$$L(P, f) := \sum_{k=1}^{n} m_k \Delta x_k$$
$$U(P, f) := \sum_{k=1}^{n} M_k \Delta x_k.$$

**Definition 7.2.2 (Upper and lower Darboux integrals)**

Let $f : [a, b] \to \mathbb{R}$ be bounded. Then, define the ***upper*** and ***lower Darboux integrals*** as follows:

$$\overline{\int_a^b} f(x)dx := \inf\{U(P, f) : P \text{ is a partiton of } [a, b]\}$$
$$\underline{\int_a^b} f(x)dx := \sup\{L(P, f) : P \text{ is a partiton of } [a, b]\}.$$

**Definition 7.2.3 (Darboux integrability)**

If $f : [a,b] \to \mathbb{R}$ is bounded, and $\underline{\int_a^b} f(x)dx = \overline{\int_a^b} f(x)dx$, then we say $f$ is ***Darboux-integrable*** ($f \in \mathcal{R}[a,b]$), and we define $\int_a^b f(x)dx$ to be the common value of the lower and upper Darboux integral.

## 7.3. Riemann integral

**Definition 7.3.1 (Riemann sum)**

Let $f : [a,b] \to \mathbb{R}$ be bounded and $P$ a partition of $[a,b]$.

For each interval $[x_{i-1}, x_i]$, select any $t_i$ in that interval. Then, the ***Riemann sum*** of $f$ under this selection of $t_i$ is

$$\sum_{i=1}^{n} f(t_i)\Delta x_i.$$

**Definition 7.3.2 (Riemann integrability)**

$f$ is ***Riemann-integrable*** ($f \in \mathcal{R}[a,b]$) iff there exists some number $I$ s.t. for every $\varepsilon > 0$, there exists a partition $P_0$ s.t. for every refinement $P$ of $P_0$ and every choice of $t_i$ on $P$, the corresponding Riemann sum is within $\varepsilon$ of $I$.

**Theorem 7.3.3**

Riemann integrability is equivalent to Darboux integrability.

## 7.4. Integrability

**Theorem 7.4.1**

Let $f : [a,b] \to \mathbb{R}$ be bounded. Then, $f$ is Darboux-integrable iff for every $\varepsilon > 0$, there exists a partition $P$ s.t. $U(P,f) - L(P,f)$.

**Theorem 7.4.2**

Let $f : [a,b] \to \mathbb{R}$ be bounded and monotonic. Then, $f \in \mathcal{R}[a,b]$.

## 7.5. Core properties of the integral

**Theorem 7.5.1 (Additivity)**

Let $a < b < c$ and $f : [a,c] \to \mathbb{R}$ be a bounded funcction. Then,

$$f \in \mathcal{R}[a,c] \iff f \in \mathcal{R}[a,b] \text{ and } f \in \mathcal{R}[b,c],$$

and if $f$ is integrable on the whole interval then

$$\int_a^c f = \int_a^b f + \int_b^c f.$$

**Theorem 7.5.2 (Linearity)**

Let $f, g \in \mathcal{R}[a,b]$, and $\alpha$ any real constant. Then,

$$\int_a^b \alpha f(x) + g(x)dx = \alpha \int_a^b f(x)dx + \int_a^b g(x)dx.$$

**Theorem 7.5.3 (Monotonicity)**

Let $f$ and $g$ be bounded functions on $[a,b]$. Assume $f(x) \leq g(x)$ for all $x \in [a,b]$. Then, $\underline{\int} f < \underline{\int} g$ and $\overline{\int} f < \overline{\int} g$. If both are in $\mathcal{R}[a,b]$ then $\int f < \int g$.

## 7.6. Additional properties of the integral

**Theorem 7.6.1**

Let $f \in [a,b]$. Let $f(x) \in [m,M]$ for all $x \in [a,b]$. Then,

$$m(b-a) \leq \int_a^b f(x)dx \leq M(b-a).$$

**Theorem 7.6.2 (Integral of a constant)**

$$\int_a^b c\,dx = c(b-a).$$

**Theorem 7.6.3 (Integral of a step function)**

Let

$$f(x) = \begin{cases} r & \text{for } x < c \\ s & \text{for } x = c, \\ t & \text{for } x > c \end{cases}$$

with $c \in [a,b]$ and $f : [a,b] \to \mathbb{R}$.

Then,

$$\int_a^b f(x)dx = r(c-a) + t(b-c).$$

**Definition 7.6.4**

$$\int_a^b f(x)dx := -\int_b^a f(x)dx.$$

**Theorem 7.6.5 (Integration by parts)**

If $u$ and $v$ are integrable and have continuous derivatives,

$$\int_a^b u(x)v'(x)dx = [u(x)v(x)]_a^b - \int_a^b u'(x)v(x)dx.$$

## 7.7. Integrals of continuous functions

**Lemma 7.7.1**

A bounded function that has only finitely many discontinuities is integrable.

**Lemma 7.7.2**

Let $f : [a,b] \to \mathbb{R}$ be bounded. Let $\{a_n\} \to a$ and $\{b_n \to b\}$ be sequences in $[a,b]$, with $a_n < b_n$ for all

$n$. Then, if $f \in \mathcal{R}[a_n, b_n]$ for all $n$, then $f \in \mathcal{R}[a, b]$ and $\int_a^b f = \lim_{n\to\infty} \int_{a_n}^{b_n} f$.

Essentially, bounded funtions need only be integrable inside the interval to be integrable on the entire interval.

**Theorem 7.7.3**

Continuous functions are integrable.

## 7.8. Fundamental theorem of calculus

**Theorem 7.8.1 (First Fundamental Theorem of Calculus)**

Let $F : [a, b] \to \mathbb{R}$ be continuous and differentiable* on $(a, b)$, with $F' \in \mathcal{R}[a, b]$. Then,

$$\int_a^b F'(x)dx = F(b) - F(a).$$

*$F$ may have finitely many points where it is not differentiable. It must be continuous over the whole interval.

**Theorem 7.8.2 (Second Fundamental Theorem of Calculus)**

Let $f \in \mathcal{R}[a, b]$. Define $F(x) := \int_a^x f$. Then, $F$ is continuous on $[a, b]$, and if $f$ is continuous at $c$, then $F$ is differentiable at $c$ and $F'(c) = f(c)$.

**Theorem 7.8.3 (u-substitution)**

Let $f : [c, d] \to \mathbb{R}$ be continuous and $g : [a, b] \subset [c, d]$ continuously differentiable. Then,

$$\int_a^b f(g(x))g'(x)dx = \int_{g(a)}^{g(b)} f(u)du.$$

Note that in this case, $u = g(x)$ and $du = g'(x)dx$.

# 8. Sequences of functions

## 8.1. Pointwise convergence

**Definition 8.1.1 (Pointwise convergence)**

Let $\{f_1, f_2, ...\}$ be a sequence of functions. Let $S$ be the set of all points $x$ for which the sequence $\{f_1(x), f_2(x), ...\}$ converges. Then, $S$ is the domain of the limit function

$$f(x) := \lim_{n\to\infty} f_n(x).$$

$\{f_n\}$ ***converges pointwise*** to $f$ on $S$.

## 8.2. Uniform convergence

**Definition 8.2.1 (Uniform convergence)**

Let $\{f_n\}$ be a sequence of functions. $\{f_n\}$ ***converges uniformly*** to $f$ on $S$ iff for every $\varepsilon > 0$ there exists $N$ s.t. $n > N$ implies $|f(x) - f_n(x)| < \varepsilon$ for all $x \in S$.

**Definition 8.2.2**

Let $\{f_n\}$ be a sequence of functions on $S$. Let $M > 0$ be a constant so that $\left|f_{n(x)}\right| \leq M$ for all $x$ and $n$. Then, $\{f_x\}$ is ***uniformly bounded*** by $M$ by $S$.

**Theorem 8.2.3**

Let $f_n \to f$ uniformly on $S$ and let each $f_n$ be bounded. Then, $f$ is bounded and the sequence $\{f_n\}$ is uniformly bounded on $S$.

**Definition 8.2.4 (Uniform norm)**

Let $f$ be a bounded function on domain $S$. Then, define the ***uniform norm*** or ***infinity norm*** of $f$ on $S$ is

$$\|f\|_\infty = \|f\|_u := \sup\{|f(x)| : x \in S\}.$$

**Theorem 8.2.5**

A sequence of bounded functions $f_n$ converges uniformly to $f$ iff

$$\lim_{n\to\infty} \|f_n - f\|_\infty = 0.$$

**Theorem 8.2.6**

Let $\{f_n\}$ be a sequence of functions. Then there exists a function $f$ s.t. $f_n \to f$ uniformly iff the sequence satisfies the ***Cauchy condition***: for every $\varepsilon > 0$ there exists $N$ s.t. if $m, n > N$, then $|f_n(x) - f_m(x)| < \varepsilon$ for all $x$.

**Definition 8.2.7**

A series converges uniformly iff its sequence of partial sums converges uniformly.

**Theorem 8.2.8 (Weierstraß $M$-test)**

Let $M_n$ be a sequence of nonnegative numbers s.t. $|f_n(x)| \leq M_n$ for all $x$ and $n$. Then, $\sum f_n$ converges uniformly if $\sum M$ converges.

## 8.3. Continuity, integrability, and differentiability of limit functions

**Theorem 8.3.1 (Interchanging limits)**

When $f_n \to f$ uniformly, then

$$\lim_{x\to a} f(x) = \lim_{x\to a} \lim_{n\to\infty} f_n(x)$$
$$= \lim_{n\to\infty} \lim_{x\to a} f_n(x) = \lim_{n\to\infty} f_n(a) = f(a).$$

**Theorem 8.3.2**

Let $\{f_n\}$ be a sequence of continuous functions $S \to \mathbb{R}$, where $S$ is some domain. If the sequence converges uniformly to a function $f$, then $f$ is also continuous.

**Theorem 8.3.3**

Let $\{f_n\}$ be a sequence of integrable functions on $[a, b]$ that converges uniformly to $f$. Then $f$ is integrable, and

$$\int_a^b f = \lim_{n\to\infty} \int_a^b f_n.$$

**Theorem 8.3.4**

Let each term of $\{f_n\}$ be differentiable at each point of $(a, b)$ and assume that these derivatives converge $f'_n \to g$ uniformly on $(a, b)$. Further, assume that there is at least one point $x_0 \in (a, b)$ s.t. $\{f_n(x_0)\}$ converges. Then,

- There is a function $f$ s.t. $f_n \to f$ uniformly on $(a, b)$, and
- For each point in $x$ in $(a, b)$, the derivative $f'(x)$ exists and equals $g(x)$.

**Theorem 8.3.5**

Upon its interval of convergence:

- The power series of a function converges uniformly to a continuous integrable
- The integral of the sum of the series is the sum of the integrals of the terms
- The derivatives of the power sums converge uniformly
- The partial sums of the series add to $f(x_0)$ at $x_0$
- Can differentiate a power series term-by-term to obtain the derivative of the function to which it converges.

# 9. Picard's theorem

**Definition 9.1 (Lipschitz in the second variable)**

$F : \mathbb{R}^2 \to \mathbb{R}$ is ***Lipschitz in the second variable*** iff there exists $L \in \mathbb{R}$ s.t.

$$|F(x, y) - F(x, z)| \leq L|y - z|$$
$$\text{for all } y, z \in J \text{ and } x \in I.$$

**Theorem 9.2 (Picard's theorem)**

Let $I, J \in \mathbb{R}$ be closed bounded intervals with $(x_0, y_0) \in I^\circ \times J^\circ$.

If $F : I \times J \to \mathbb{R}$ is continuous and Lipschitz in the second variable, then there exists $h > 0$ s.t. $[x_0 - h, x_0 + h] \subset I$ and a unique differentiable function $f : [x_0 - h, x_0 + h] \to J$ s.t.

$$f'(x) = F(x, f(x)) \qquad f(x_0) = y_0.$$

**In short,** if $F$ is continuous and Lipschitz in the second variable, the initial value problem has a unique solution on some interval around $x_0$.

The unique solution can be constructed via the following procedure:

1. Define $f_0(x) := y_0$.
2. For each $k \in \mathbb{N}^+$, define the ***kth Picard iterate***:

$$f_k(x) := y_0 + \int_{x_0}^x F(t, f_{k-1}(t))dt$$

3. The sequence of Picard iterates converges uniformly to the solution $f$.

# 10. Lebesgue integrability condition

## 10.1. Oscillation

**Definition 10.1.1 (Oscillation)**

Let $f : \mathbb{R} \to \mathbb{R}$ and $S \subset \mathbb{R}$. The ***oscillation*** of $f$ on $S$ is defined as

$$\omega_f(S) := \sup\{f(S)\} - \inf\{f(S)\}.$$

If the supremum or infimum does not exist, it is defined as $+\infty$.

The ***oscillation*** of $f$ at the point $x \in S$ is defined as

$$\lim_{\varepsilon\to 0} \omega_f(S \cap (x - \varepsilon, x + \varepsilon)).$$

**Theorem 10.1.2**

$f$ is continuous at $x_0$ iff the oscillation of $f$ at $x_0$ is zero.

## 10.2. Measure

**Definition 10.2.1 (Measure)**

A ***measure*** is a function that takes a subset $S$ of the reals, and returns its "size" denoted $|S|$, following the following properties:

- $0 \leq |S| \leq \infty$.
- $|\varnothing| = 0$.
- $S' := \{x + a : x \in S\} \implies |S'| = |S|$.
- $|(0, 1)| = |[0, 1]| = 1$.
- ***Countably additive***: If $\{S_i\}$ is a countable collection of disjoint sets, $|\cup S_i| = \sum_i |S_i|$ .

### 10.2.1. Lebesgue outer measure

**Definition 10.2.2 (Lebesgue outer measure)**

Cover the set with a countable number of disjoint open intervals, and sum their totals. Take the infimum of all such coverings.

$$\lambda^*(S) := \inf\left\{\sum (b_i - a_i)\{(a_i, b_i)\}_{i\in\mathbb{Z}^+} \text{ is such a cover}\right\}.$$

**Theorem 10.2.3**

- Every countable subset of $\mathbb{R}$ has [Lebesgue outer] measure 0.
- A subset of a measure 0 set also has measure 0.
- Any countable union of measure 0 sets has measure 0.

## 10.3. Lebesgue integrability condition

**Theorem 10.3.1 (Lebesgue Integrability Condition)**

$f : [a, b] \to \mathbb{R}$ is Riemann-integrable iff it is bounded and its set of discontinuities has measure 0.

**Corollary 10.3.1.1**

Let $f : [a, b] \to [c, d]$ be integrable and $g : [c, d] \to \mathbb{R}$ continuous. Then, $g \circ f$ is integrable.

# 11. Counterexamples & Constructions

## 11.1. Pathological Functions

**P1. Dirichlet function**: $D(x) = 1$ if $x \in \mathbb{Q}$, 0 if $x \notin \mathbb{Q}$. *Properties*: bounded; nowhere continuous (every interval contains both rationals and irrationals); NOT integrable ($U(P, D) = 1$, $L(P, D) = 0$ for any partition $P$); can be written as $D(x) = \lim_{m\to\infty}(\lim_{n\to\infty} \cos^{2n}(m!\pi x))$.

**P2. Thomae's function** (Modified Dirichlet): $f\left(\frac{p}{q}\right) = \frac{1}{q}$ (lowest terms), $f(\text{irrational}) = 0$. *Properties*: bounded; continuous at every irrational (hence almost everywhere); discontinuous at every rational; Riemann integrable with $\int_0^1 f = 0$; discontinuities are countable (measure 0). Shows the Lebesgue Integrability Condition in action.

**P3.** $f(x) = x \sin\left(\frac{1}{x}\right)$, $f(0) = 0$. *Properties*: continuous everywhere; uniformly continuous on $[0, 1]$; NOT Lipschitz at 0 (oscillates arbitrarily steeply); NOT differentiable at 0.

**P4.** $f(x) = x^2 \sin\left(\frac{1}{x}\right)$, $f(0) = 0$. *Properties*: differentiable everywhere; $f'(0) = 0$; $f'(x) = 2x \sin\left(\frac{1}{x}\right) - \cos\left(\frac{1}{x}\right)$ for $x \neq 0$; $f' \not\to 0$ as $x \to 0$, so $f'$ is NOT continuous at 0 ($f$ is differentiable but $f' \notin C^1$).

**P5.** $f_{n(x)} = x^n$ on $[0, 1]$. Pointwise limit: $f(x) = 0$ for $x < 1$, $f(1) = 1$ (discontinuous). Convergence is NOT uniform: $\|f_n - f\|_\infty = 1$ for all $n$. Shows the uniform limit of continuous functions need not hold under pointwise convergence.

**P6. Weierstraß function**: $W(x) = \sum_{n=0}^{\infty} \left(\frac{1}{2}\right)^n \cos(3^n \pi x)$. Continuous everywhere, nowhere differentiable ($\|f_n - f\|_\infty \to 0$ by the Weierstraß $M$-test, but derivatives blow up). Limit of smooth functions can fail to be smooth.

**P7.** $f(x) = \sqrt{x}$ on $[0, 1]$. Uniformly continuous (compact domain and continuous). NOT Lipschitz at 0: $\frac{|\sqrt{x}-0|}{|x-0|} = \frac{1}{\sqrt{x}} \to \infty$. Shows uniform continuity $\not\Rightarrow$ Lipschitz.

**P8.** $f(x) = \frac{1}{x}$ on $(0, 1)$. Continuous. NOT uniformly continuous ($\delta$ must depend on position; Cauchy sequences map to non-Cauchy sequences). NOT bounded. Shows $f : (a, b) \to \mathbb{R}$ is uniformly continuous if and only if it extends continuously to $[a, b]$.

**P9.** $f(x) = |x|$. Uniformly continuous; Lipschitz (constant $K = 1$); NOT differentiable at 0. Shows Lipschitz $\not\Rightarrow$ differentiable.

**P10.** If $|f(x) - f(y)| \leq (x - y)^2$ for all $x, y$, then $f'(x) = \lim_{h\to 0} \frac{f(x+h)-f(x)}{h} = 0$ everywhere, so $f$ is constant.

**P11.** The alternating harmonic series $\sum \frac{(-1)^n}{n}$ converges (by the alternating series test) but NOT absolutely (since $\sum \frac{1}{n}$ diverges). Any real number can be achieved by rearrangement (Riemann Rearrangement Theorem).

## 11.2. Property Constructions (No Proof)

- **Continuous, not uniformly continuous**: $f(x) = x^2$ on $\mathbb{R}$; or $\sin\left(\frac{1}{x}\right)$ on $(0, 1]$.
- **Uniformly continuous, not Lipschitz**: $f(x) = \sqrt{x}$ on $[0, 1]$.
- **Lipschitz, not differentiable**: $f(x) = |x|$ (constant $K = 1$).
- **Pointwise convergent, not uniformly convergent**: $f_{n(x)} = x^n$ on $[0, 1]$.
- **Uniformly convergent, but derivatives may not converge**: $f_{n(x)} = \frac{\sin(nx)}{n} \to 0$ uniformly, but $f_{n'}(x) = \cos(nx) \not\to 0$.
- **Bounded, not integrable**: Dirichlet function $D(x)$.
- **Integrable, not continuous (even with dense discontinuities)**: Thomae's function.
- **Integrable $f$, injective $g$, but $f \circ g$ not integrable**: Take $f$ = Thomae (discontinuous on $\mathbb{Q}$) and $g$ a bijection mapping irrationals to rationals (e.g. Cantor-based); $f \circ g$ is 0 on irrationals, $\frac{1}{q}$ on images of rationals — can be made non-integrable.
- **Sequence with lim sup $\neq$ lim inf**: $\{(-1)^n\}$: $\limsup = 1$, $\liminf = -1$.
- **Limit comparison test failure**: $\sum \frac{1}{n}$ and $\sum \frac{1}{n^2}$: ratio $\to \infty$, both diverge/converge differently — shows the limit comparison test requires $0 < L < \infty$.

## 11.3. Quick Reference: Implications

- Lipschitz $\Rightarrow$ Uniformly continuous $\Rightarrow$ Continuous
- Differentiable $\Rightarrow$ Continuous
- Uniformly convergent $\Rightarrow$ Pointwise convergent
- Absolutely convergent $\Rightarrow$ Convergent
- Closed + bounded $\iff$ Sequentially compact
- Continuous $\Rightarrow$ Integrable

None of these reverse in general.